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* Twitter - QPS & Storage Estimation :-

⇓

→ Question Related to Back of the Envelope Estimation

Q^n :- Estimate Twitter's QPS (Query per second) and storage Requirements

Note :- Following numbers are taken for example only.

★ ASSUMPTIONS :-

- Monthly active users = 300 million
- Daily users = 50%
- Tweets per day on Avg. per user = 2
- Tweets containing media = 10%
- Tweets stores for = 5 years

★ Estimation :-

QPS (Query per second) estimate :-

⇒ daily active users (DAV) = 50% of 300 million

= 150 million

⇒ Tweets QPS = $\frac{150 \text{ million} * 2 \text{ tweets}}{24 \text{ h} * 3600 \text{ sec}}$

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$$= \frac{300,000,000}{86,400} \simeq \underbrace{3,472.222}_{\text{tweets per second}}$$

$$= \sim 3500$$

$\Rightarrow$ Peek QPS $= 2 * QPS = \sim 70,00$
(100 % users)

★ STORAGE Requirements Estimation :-

Note :- we will only estimate media storage here → photo/video

$\Rightarrow$ Average tweet size :

- tweet-id = 64 bytes
- text = 140 bytes
- Media = 1 MB

$\Rightarrow$ Media Storage $= \underbrace{(\text{per day active users})}_{150 \text{ million}} * \underbrace{(\text{no. of tweets per users})}_{2} * \underbrace{(\text{10\% contain media})}_{10\%} * \underbrace{(\text{size of media})}_{1 \text{ MB}}$

$\because 1 TB = 10^6 MB$

$= 150 \text{ million} * 2 * 10\% * 1 MB = 30 \times 10^6$

$= \boxed{30 \text{ TB per day}}$

$\Rightarrow$ 5 year Media storage $= (30 TB) * (365) * 5$

$= 54,750 TB$

$\boxed{\approx 55 PB}$

$\because 1 PB = 1000 TB$

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Interview tips :-

- Rounding & Approximation → 99989 / 9.2 ✗ → 10,000 / 10 ✓
- Always write down your Assumptions.
- label your units

  size 5 ✗

  size 5MB ✓

- commonly asked Questions :- QPS, Peek QPS, storage Cache, no. of servers etc.